

Formalizing Privacy in Decentralized Identity: A Provably Secure Framework with Minimal Disclosure

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Abstract—This paper presents a formal framework for enhancing privacy in decentralized identity (DID) systems, resolving the inherent conflict between blockchain verifiability and the principle of minimal data disclosure. At its core, we introduce a provably secure cryptographic protocol that leverages attribute commitments on-chain and zero-knowledge proofs for off-chain validation. This approach allows users to demonstrably prove the validity of predicates about their attributes without revealing the underlying sensitive values.

We formally define the security and privacy requirements for such a system—including consistency, attribute-based indistinguishability, and predicate-based indistinguishability—within a semi-honest adversarial model. We then construct a concrete scheme that realizes these properties under standard cryptographic assumptions. The proposed architecture is designed for full backward compatibility with W3C DID standards, ensuring practical deployability. Security analysis provides rigorous, provable guarantees, while performance evaluation confirms the efficiency of the core cryptographic operations, supporting its use in resource-constrained environments. This work establishes a foundational and analyzable basis for building decentralized identity systems where both accountability and user privacy are essential.

Index Terms—Decentralized Identity, Minimal Disclosure, Zero-Knowledge Proofs, Pedersen Commitments, Formal Security Model

I. INTRODUCTION

Decentralized Identity (DID) systems[1], [2], [3], [4], [5] have emerged as a promising alternative to traditional identity management, addressing critical issues of centralized control and single points of failure[6], [7]. Built on blockchain technology[8], [9], [10], [11], [12], [13], DIDs enable users to own and control their digital identities without relying on intermediaries[14], [15]. The standardization efforts, notably the W3C DID [16] specification, have paved the way for interoperable solutions. However, the architectural shift to decentralization introduces novel and profound privacy challenges that existing systems and standards are ill-equipped to handle formally.

While the W3C Verifiable Credentials standard rightly cautions against on-chain storage of sensitive data, the mechanisms it suggests for selective disclosure, such as the omission of entire claims using Merkle proofs or derived credentials, are

primarily syntactic and lack rigorous cryptographic privacy guarantees [16]. Conventional approaches, such as storing hashes or specific predicates on-chain [17], [18], [19], either fail to achieve minimal disclosure or leak information through the predicates themselves. A core issue is the absence of a formal security model that precisely captures the privacy guarantees required in a decentralized, publicly verifiable setting. This gap makes it impossible to reason rigorously about whether these systems protect user data against even passive, semi-honest adversaries.

The inability to enforce the principle of minimal disclosure [20], [21], [22], [23], [24] is particularly acute. Proving an attribute like $\text{age} \geq 18$ should not necessitate revealing one's exact birth date. While zero-knowledge proofs (ZKPs) are a promising cryptographic tool to bridge this gap, their ad-hoc application without a formal foundation leaves systems vulnerable to design flaws and subtle attacks. Therefore, the critical research question we address is: How can we formally define and provably achieve minimal disclosure for decentralized identity?

A. Related Work

Previous research on privacy in DID systems has largely focused on engineering pragmatic solutions, often at the expense of formal rigor and provable guarantees.

The work of [17] enhances privacy by obfuscating attribute-owner linkages through hashing. However, it ultimately requires full attribute disclosure during authentication. This scheme lacks a formal privacy model, and its security against even semi-honest verifiers who observe the blockchain is unclear.

The approach of [18], which stores predicates on-chain, represents a step towards minimal disclosure but introduces significant inefficiencies and metadata leakage. More critically, it does not formalize the privacy notion it seeks to achieve. The published predicates themselves form a side channel, and the work does not provide a security proof demonstrating that the system reveals nothing beyond the predicate's truth value.

Systems like [25] rely on permissioned blockchains and Secure Multi-Party Computation to protect on-chain data.

While this provides a form of privacy, it fundamentally departs from the public verifiability model of mainstream DIDs and substitutes technical guarantees with administrative access control. This line of work does not address the core challenge of achieving privacy on a transparent ledger.

In summary, a common shortcoming across these approaches is the lack of a comprehensive formal framework. None present a unified set of security definitions—such as indistinguishability for committed attributes and for predicate proofs—that would allow for rigorous analysis and comparison. Our work directly addresses this gap by providing formal model for privacy in decentralized identity that encompasses both credential issuance and presentation, alongside constructions that are provably secure under standard assumptions.

B. Contributions

This work establishes a **formal framework** for decentralized identity management that achieves provable privacy guarantees under the principle of minimal disclosure. The framework comprises the following foundational contributions:

- **A Formal Security Model for Privacy in Decentralized Identity.** We develop a comprehensive formalization of privacy for blockchain-based DID systems in the presence of semi-honest adversaries. Our model captures the desired security properties through three core definitions:
 - **Consistency**, which guarantees the cryptographic soundness of the credential issuance protocol, ensuring an adversary cannot generate a valid credential for a false attribute value.
 - **Attribute-Based Indistinguishability**, which formalizes the hiding property of on-chain data, ensuring that the public commitments reveal no information about the underlying sensitive attributes.
 - **Predicate-Based Indistinguishability**, which captures the principle of minimal disclosure during credential presentation, ensuring that a zero-knowledge proof reveals only the truth of a predicate and nothing more about the specific attribute value.
- **A Provably Secure Construction.** We instantiate our formal model with a concrete and efficient cryptographic protocol. This scheme uniquely integrates off-chain attribute storage with on-chain Pedersen commitments [26], [27] to ensure attribute confidentiality, while leveraging efficient zero-knowledge proofs [28], [29], [30], [31] for predicate verification. Crucially, we provide **rigorous security proofs**, demonstrating that our construction satisfies all defined properties under standard cryptographic assumptions, namely the hardness of the Elliptic Curve Discrete Logarithm Problem (ECDLP) and the anonymity of the underlying ring signature scheme.
- **Practical Validation without Compromising Theoretical Rigor.** We demonstrate that a provably secure foundation does not preclude practical deployability. Our construction is designed with efficiency in mind, featuring linear-time operations and constant-sized on-chain

records. An experimental evaluation confirms the computational feasibility of the core cryptographic operations, even on resource-constrained devices. Furthermore, the architecture is fully backward-compatible with W3C DID standards [16], ensuring that our theoretical advances can be integrated into the existing ecosystem.

II. PROBLEM STATEMENT AND SYSTEM MODEL

A. Problem Statement

We address the foundational challenge of designing a decentralized identity system that achieves both *public verifiability* and *strong attribute privacy*. The core problem can be stated as enabling a user to convincingly demonstrate possession of credentials with certain properties to a service provider, while cryptographically enforcing the principle of minimal disclosure. This must be achieved under the following constraints and security assumptions:

- **Adversarial Model:** All participants except for the Identity Providers (IdPs) are modeled as *semi-honest* (i.e., honest-but-curious). They faithfully follow the protocol specification but may attempt to learn additional information about a user's attributes through passive observation and analysis of all public and protocol-derived data. IdPs are trusted to correctly generate and update policy commitments.
- **Verifiability Constraint:** Identity documents (including DID records and credential status) must be anchored on a *publicly accessible* blockchain. This ensures global verifiability, non-repudiation, and prevents single points of failure inherent in centralized systems.
- **Privacy Requirement:** Sensitive user *privacy attributes* (e.g., age, salary, or affiliations) must remain confidential against all other parties. A Service Provider (SP) should be able to verify *only* the satisfaction of a specific predicate θ (e.g., “Age ≥ 18 ”) without learning the underlying attribute value v itself. Any solution must prevent leakage of v through the credential's on-chain representation or the interactive proof protocol.

The central goal is to construct a cryptographic protocol that fulfills these requirements and for which the stated privacy guarantees can be formally defined and rigorously proven.

B. System Model

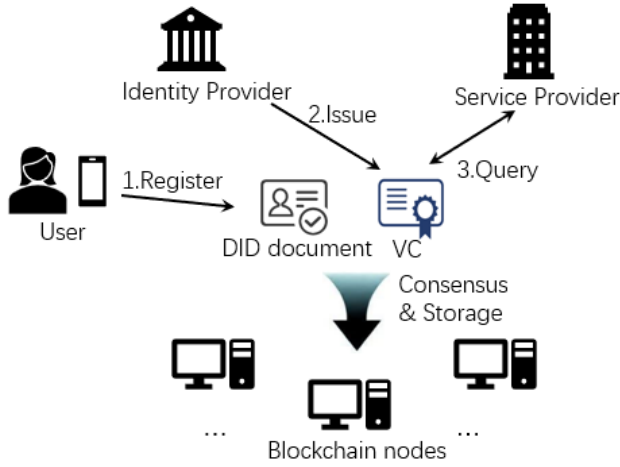


Fig. 1. System architecture of DIMSEPP

To address the problem defined in Section II-A, we propose a concrete system architecture, referred to as **DIMSEPP** (Decentralized Identity Management System with Enhanced Privacy Protection). This architecture provides the structural foundation for instantiating our formal privacy framework and enables cryptographic protocols that enforce minimal disclosure.

The DIMSEPP architecture (Fig. 1) extends standard DID protocols with cryptographic privacy enhancements. The system operates through three sequential phases that form the complete identity lifecycle.

In the **registration phase**, users generate DID documents containing public keys and register them via blockchain transactions. These documents are validated through the blockchain's consensus mechanism and permanently stored across the distributed network of blockchain nodes, establishing a globally verifiable cryptographic identity.

In the **credential issuance phase**, users request verifiable credentials (VCs) from Identity Providers (IdPs). This involves off-chain verification where the IdP validates the user's attribute evidence (e.g., government-issued documents), a critical privacy-preserving step that occurs outside the public ledger. Upon successful verification, the IdP issues a signed VC that contains cryptographically protected attribute commitments rather than plain values. This VC is then anchored on the blockchain through the same consensus process, creating an immutable link between the user's DID and their attested attributes.

In the **service access phase**, users generate zero-knowledge proofs (ZKPs) demonstrating that their committed attributes satisfy specific predicates (e.g., proving $\text{age} \geq 18$) without revealing the actual attribute values. Service Providers (SPs) then query the blockchain (represented by the bi-directional arrow in Fig. 1) to fetch the relevant VC using the user's DID, and verify these ZKPs against this on-chain credential.

Through this process, SPs learn only the truth value of the predicate while preserving full attribute confidentiality.

For credential lifecycle management, IdPs may revoke compromised credentials via signed revocation transactions, while users initiate updates by submitting new attribute commitments with ZKPs of validity. These operations follow standard DID protocols for revocation and updates, ensuring compatibility with existing ecosystems.

We assume the standard cryptographic practice for handling verifiable credentials. Specifically, a VC is considered *valid* if it carries a valid digital signature from its issuer (IdP). The verification of a VC is therefore reduced to standard signature verification using the IdP's public key, which is embedded in its own DID document on the blockchain. This process is orthogonal to the novel cryptographic operations presented in this work and is thus omitted from the core algorithm descriptions.

The core privacy mechanism of DIMSEPP relies on a cryptographic commitment scheme. We model this with an abstract interface.

A **ProtectedAttribute** for a value v is generated by an algorithm $\text{Protect}(pp, v, \text{aux}) \rightarrow (C, \pi_{\text{open}})$, where pp is public parameters.

- C is the protected value (commitment).
- π_{open} is a zero-knowledge proof that certifies the correct formation of C . Specifically, it proves knowledge of an auxiliary value aux such that C is a valid commitment to v using aux .

A corresponding verification algorithm $\text{VerifyOpen}(C, v, \pi_{\text{open}}) \rightarrow \{0, 1\}$ is used to verify the proof. The scheme must provide:

- **Hiding:** C reveals no information about v .
- **Binding:** It is computationally infeasible for an adversary to find a commitment C and two distinct pairs (v, π_{open}) and (v', π'_{open}) with $v \neq v'$, such that both $\text{VerifyOpen}(C, v, \pi_{\text{open}}) = 1$ and $\text{VerifyOpen}(C, v', \pi'_{\text{open}}) = 1$ hold.
- **Zero-Knowledge:** π_{open} reveals nothing about aux beyond the fact that the statement is true.

The system's core operations (excluding revocation/update) are implemented through six algorithms that support the three operational phases described above. Specifically:

- **Setup** provides foundational parameters for the entire system
- **CreateDIDD** implements the registration phase
- **ApplyCred** and **GenCred** implement the credential issuance phase
- **ApplyServ** and **VerifyServ** implement the service access phase

These algorithms utilize the abstract ProtectedAttribute interface as follows:

- 1) **Setup**(1^λ) $\rightarrow pp$: Setup is executed by some party. Takes a security parameter λ , Setup generates pp and publishes them, which are available to all parties. The setup is done only once.

- 2) **CreateDIDD**(pp) $\rightarrow (didd, pk, sk)$: This is executed by users. Take pp as input, **CreateDIDD** generates key pair (pk, sk) and DID document $didd$ containing pk .
- 3) **ApplyCred**(pp, v, aux) $\rightarrow (C, \pi_{open})$: The user executes **Protect** with given pp, v and aux as input, which produces C and π_{open} . Then the algorithm returns the tuple (C, π_{open}) .
- 4) **GenCred**($pp, didd, attr, v, C, \pi_{open}, \text{Proof}_v$) $\rightarrow VC$: With $pp, didd, attr, v, C, \pi_{open}$ and Proof_v (valid external evidence for v) as input, IdP performs **GenCred** to generate verifiable credential VC .
- 5) **ApplyServ**(pp, v, aux, θ) $\rightarrow \pi_\theta$: With pp, v, aux and predicate θ specifying a condition on v (e.g., $\theta := "v \geq 18"$ for an age attribute) as input, user performs **ApplyServ** to generate ZKP π_θ .
- 6) **VerifyServ**($pp, C, \theta, \pi_\theta$) $\rightarrow \{0, 1\}$: With $pp, C, \theta, \pi_\theta$ as input, SP executes **VerifyServ** to determine whether to grant the service or not.

C. Definitions

Definition 1 (Completeness) A DIMSEPP scheme $\Pi = (\text{Setup}, \text{CreateDIDD}, \text{ApplyCred}, \text{GenCred}, \text{ApplyServ}, \text{VerifyServ})$ is *complete* if for any polynomial-size ledger sampler $\mathcal{S}$ and security parameter λ , the following incompleteness advantages are negligible in λ . These advantages capture the probability that an *honest user* following the protocol specification fails to achieve the intended outcome—either in *obtaining a verifiable credential* from a trusted issuer or in *accessing a service* for which the user satisfies the predicate.

- **VC Generation Completeness:** $\text{Adv}_{\Pi, \mathcal{S}}^{\text{VC-COMP}} = \Pr[\text{VC-COMP}(\Pi, \mathcal{S}, \lambda) = 1]$
- **Service Access Completeness:** $\text{Adv}_{\Pi, \mathcal{S}}^{\text{SERV-COMP}} = \Pr[\text{SERV-COMP}(\Pi, \mathcal{S}, \lambda) = 1]$

The experiments proceed as follows:

- **VC-COMP**($\Pi, \mathcal{S}, \lambda$): This experiment tests the completeness of the credential issuance protocol. It models a scenario where an honest user, possessing valid external evidence Proof_v for an attribute value v , correctly computes a commitment C and a proof π_{open} . The experiment outputs 1 if the Identity Provider, upon verifying the proof and the evidence, *fails to generate a valid verifiable credential* VC . A non-negligible advantage here would indicate a fundamental flaw in the protocol's design that prevents honest users from obtaining credentials.
 - 1) Challenger $\mathcal{C}$ generates $pp \leftarrow \text{Setup}(1^\lambda)$ and sends it to $\mathcal{S}$.
 - 2) $\mathcal{S}$ submits to $\mathcal{C}$:
 - Ledger state L containing $didds$ and VCs .
 - Attribute tuple $(attr, v, aux, \text{Proof}_v)$.
 - 3) $\mathcal{C}$ generates $(C, \pi_{open}) \leftarrow \text{ApplyCred}(pp, v, aux)$.
 - 4) $\mathcal{C}$ outputs 1 if **GenCred**($pp, didd, attr, v, C, \pi_{open}, \text{Proof}_v$) fails; else 0.
- **SERV-COMP**($\Pi, \mathcal{S}, \lambda$): This experiment tests the completeness of the service access protocol.
 - 1) $\mathcal{C}$ initializes as in VC-COMP.

- 2) $\mathcal{S}$ submits to $\mathcal{C}$:
 - L containing $didds$ and VCs .
 - Tuple (v, aux, θ) .
- 3) $\mathcal{C}$ generates $\pi_\theta \leftarrow \text{ApplyServ}(pp, v, aux, \theta)$.
- 4) $\mathcal{C}$ outputs 1 if either:
 - $\text{VerifyServ}(pp, C, \theta, \pi_\theta) = 0$ when v satisfies θ , or
 - $\text{VerifyServ}(pp, C, \theta, \pi_\theta) = 1$ when v violates θ

Definition 2 (Security) A DIMSEPP scheme Π is *secure* if it satisfies *consistency*, *attribute-based indistinguishability* and *predicate-based indistinguishability*.

Each of the properties is defined by an experiment involving a DIMSEPP oracle $\mathcal{O}^{\text{DIMSEPP}}$ that receives and answers queries from an adversary $\mathcal{A}$. The queries are proxied via $\mathcal{C}$ that performs the experiment-specific sanity checks.

$\mathcal{O}^{\text{DIMSEPP}}$ maintains state $(pp, L, \text{KEYS}, \text{VR})$, initialized with $pp \leftarrow \text{Setup}(1^\lambda)$ and empty sets $L, \text{KEYS}, \text{VR}$. Here, L is designed to be **publicly readable** by all parties. The oracle processes the following query types:

- $Q = (\text{CreateDIDD})$:
 - Compute $(didd, pk, sk) \leftarrow \text{CreateDIDD}(pp)$
 - Update $\text{KEYS} \leftarrow \text{KEYS} \cup \{(pk, sk)\}$
 - Update $L \leftarrow L \cup didd$
- $Q = (\text{ApplyCred}, didd, attr, v, aux)$:
 - Compute $(C, \pi_{open}) \leftarrow \text{ApplyCred}(pp, v, aux)$
 - Update $\text{VR} \leftarrow \text{VR} \cup (didd, attr, v, aux, C)$
 - Return (C, π_{open})
- $Q = (\text{GenCred}, didd, attr, v, C, \pi_{open}, \text{Proof}_v)$:
 - Verify $didd \in L$; if fails, abort
 - Validate external evidence Proof_v ; if invalid, abort. (This step represents the IdP's off-chain verification process, which confirms the accuracy of the $attr$ value v , e.g., verifying a passport signature to validate age.)
 - If $\text{VerifyOpen}(C, v, \pi_{open}) = 0$, abort (In our concrete construction, this is $\text{Schnorr.Verify}(C - [v]G, H, \pi_r)$.)
 - Compute $VC \leftarrow \text{GenCred}(pp, didd, attr, v, C, \pi_{open}, \text{Proof}_v)$
 - Update $L \leftarrow L \cup \{VC\}$.
- $Q = (\text{ApplyServ}, didd, \theta)$:
 - Verify $didd \in L$; if fails, abort
 - Retrieve (v, aux, C) associated with $(didd, \theta.attr)$ from VR ; if fails, abort
 - If v doesn't satisfy θ , abort
 - Compute $\pi_\theta \leftarrow \text{ApplyServ}(pp, v, aux, \theta)$.
 - Return π_θ (does not update state).
- $Q = (\text{VerifyServ}, didd, \theta, \pi_\theta)$:
 - Verify $didd \in L$; if fails, abort.
 - Retrieve C from $L.VC$ associated with $(didd, \theta.attr)$; if fails, abort.
 - Compute $b \leftarrow \text{VerifyServ}(pp, C, \theta, \pi_\theta)$.
 - Return b (does not update state).

Definition 2.1 (Consistency) A DIMSEPP scheme Π satisfies *consistency* if for all Probabilistic Polynomial-Time (PPT) adversaries $\mathcal{A}$ and security parameter λ , the advantage

$\text{Adv}_{\Pi, \mathcal{A}}^{\text{CONS}}(\lambda)$ is negligible:

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{CONS}}(\lambda) := \Pr[\text{CONS}(\Pi, \mathcal{A}, \lambda) = 1]$$

The experiment $\text{CONS}(\Pi, \mathcal{A}, \lambda)$ proceeds as:

- 1) **Initialization:**
 - $\mathcal{C}$ initializes $\mathcal{O}^{\text{DIMSEPP}}$ and gives pp to $\mathcal{A}$
- 2) **Query Phase:**
 - $\mathcal{A}$ makes adaptive queries to $\mathcal{O}^{\text{DIMSEPP}}$
- 3) **Challenge Phase:**
 - $\mathcal{A}$ outputs $(v^*, v'^*, \text{aux}'^*, C^*, \pi_{\text{open}}^*)$ with:
 - $(C^*, _) \leftarrow \text{Protect}(pp, v'^*, \text{aux}'^*)$
 - $\text{VerifyOpen}(C^*, v^*, \pi_{\text{open}}^*) = 1$
 - $v^* \neq v'^*$
- 4) **Output Phase:**
 - $\mathcal{C}$ outputs 0 unless all of the following hold:
 - $\text{VerifyOpen}(C^*, v^*, \pi_{\text{open}}^*) = 1$
 - $v^* \neq v'^*$
 - $\mathcal{C}$ attempts $\text{GenCred}(pp, \text{didd}, \text{attr}, v^*, C^*, \pi_{\text{open}}^*, \text{Proof}_v^*)$ and outputs 1 if valid VC^* is generated, else 0

Consistency ensures the *cryptographic soundness* of the credential issuance protocol. It guarantees that an adversary cannot create a valid credential for a false attribute value. This property is fundamental to the system's integrity.

Definition 2.2 (Attribute-based Indistinguishability) A DIMSEPP scheme Π achieves *attribute-based indistinguishability* if for PPT adversaries $\mathcal{A}$ and security parameter λ , the advantage $\text{Adv}_{\Pi, \mathcal{A}}^{\text{AIND}}(\lambda)$ is negligible:

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{AIND}}(\lambda) := |\Pr[\text{AIND}(\Pi, \mathcal{A}, \lambda) = 1] - \frac{1}{2}|$$

The experiment $\text{AIND}(\Pi, \mathcal{A}, \lambda)$ proceeds as:

- 1) **Initialization:**
 - $\mathcal{C}$ generates $pp \leftarrow \text{Setup}(1^\lambda)$
 - $\mathcal{C}$ initializes $\mathcal{O}^{\text{DIMSEPP}}$ with pp
 - $\mathcal{C}$ provides pp to $\mathcal{A}$ and oracle access to $\mathcal{O}^{\text{DIMSEPP}}$
- 2) **Query Phase 1:**

$\mathcal{A}$ may adaptively issue queries of type Q to $\mathcal{O}^{\text{DIMSEPP}}$ through $\mathcal{C}$, where Q can be:

 - (CreateDiDD)
 - (ApplyCred, $\text{didd}, \text{attr}, v, \text{aux}$)
 - (GenCred, $\text{didd}, \text{attr}, v, C, \pi_{\text{open}}, \text{Proof}_v$)

$\mathcal{O}^{\text{DIMSEPP}}$ processes these queries according to their predefined specifications.
- 3) **Challenge Phase:**
 - $\mathcal{A}$ submits (v_0, v_1) where $v_0 \neq v_1$
 - $\mathcal{C}$ selects $b \leftarrow \{0, 1\}$
 - $\mathcal{C}$ invokes $\mathcal{O}^{\text{DIMSEPP}}$ to compute $(C_b, \pi_{\text{open}}) \leftarrow \text{ApplyCred}(pp, v_b, \text{aux})$, where aux is selected by $\mathcal{C}$.
 - $\mathcal{C}$ returns C_b to $\mathcal{A}$
- 4) **Query Phase 2:**
 - $\mathcal{A}$ continues queries to $\mathcal{O}^{\text{DIMSEPP}}$ through $\mathcal{C}$

- **Restriction:** No direct queries about (v_b, C_b)

5) Output Phase:

- $\mathcal{A}$ outputs $b^* \in \{0, 1\}$ as its guess to b
- $\mathcal{C}$ outputs 1 if $b = b^*$, else 0

The DIMSEPP oracle $\mathcal{O}^{\text{DIMSEPP}}$ maintains all system state (including ledger L , key store **KEYS**, and credential records **VR**), while $\mathcal{C}$ acts as a relay between $\mathcal{A}$ and $\mathcal{O}^{\text{DIMSEPP}}$. All cryptographic computations are performed by $\mathcal{O}^{\text{DIMSEPP}}$ to properly model system execution.

Attribute-Based Indistinguishability (AIND) ensures that the Pedersen commitment C within a verifiable credential stored on-chain reveals no information about the sensitive attribute value v it commits to. This formalizes the *hiding* property of the commitment [26] in our system context, guaranteeing confidentiality against any adversary who observes the public blockchain.

Definition 2.3 (Predicate-Based Indistinguishability) A DIMSEPP scheme Π satisfies *predicate-based indistinguishability* if for all PPT adversaries $\mathcal{A}$ and security parameter λ , the advantage $\text{Adv}_{\Pi, \mathcal{A}}^{\text{PIND}}(\lambda)$ is negligible, where the experiment $\text{PIND}(\Pi, \mathcal{A}, \lambda)$ proceeds as:

1) Initialization:

- $\mathcal{C}$ generates $pp \leftarrow \text{Setup}(1^\lambda)$
- $\mathcal{C}$ initializes $\mathcal{O}^{\text{DIMSEPP}}$ with pp
- $\mathcal{C}$ provides pp to $\mathcal{A}$ and oracle access to $\mathcal{O}^{\text{DIMSEPP}}$

2) Query Phase 1:

$\mathcal{A}$ may adaptively issue queries of type Q to $\mathcal{O}^{\text{DIMSEPP}}$ through $\mathcal{C}$, where Q can be:

- (CreateDiDD)
- (ApplyCred, $\text{didd}, \text{attr}, v, \text{aux}$)
- (GenCred, $\text{didd}, \text{attr}, v, C, \pi_{\text{open}}, \text{Proof}_v$)
- (ApplyServ, didd, θ)
- (VerifyServ, $\text{didd}, \theta, \pi_\theta$)

$\mathcal{O}^{\text{DIMSEPP}}$ processes these queries according to their predefined specifications.

3) Challenge Phase:

- $\mathcal{A}$ submits (v_0, v_1, θ) where:
 - $v_0 \neq v_1$
 - Both v_0 and v_1 satisfy the θ
- $\mathcal{C}$ selects $b \leftarrow \{0, 1\}$
- $\mathcal{C}$ selects aux , invokes $\mathcal{O}^{\text{DIMSEPP}}$ to compute:

$$(C_b, \pi_{\text{open}}) \leftarrow \text{ApplyCred}(pp, v_b, \text{aux}),$$

$$\pi_\theta \leftarrow \text{ApplyServ}(pp, v_b, \text{aux}, \theta)$$

- $\mathcal{C}$ returns (C_b, π_θ) to $\mathcal{A}$

4) Query Phase 2:

- $\mathcal{A}$ continues queries to $\mathcal{O}^{\text{DIMSEPP}}$ through $\mathcal{C}$
- **Restriction:** No queries about (v_b, C_b, π_θ)

5) Output Phase:

- $\mathcal{A}$ outputs $b^* \in \{0, 1\}$ as its guess for b
- $\mathcal{C}$ outputs 1 if $b = b^*$, else 0

The advantage is defined as:

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{PIND}}(\lambda) := \left| \Pr[\text{PIND}(\Pi, \mathcal{A}, \lambda) = 1] - \frac{1}{2} \right|$$

Predicate-Based Indistinguishability (PIND) ensures that a zero-knowledge proof π_θ revealed during service access proves only the truth of the predicate θ (e.g., “age ≥ 18 ”) and nothing more about the specific value v . This captures the principle of *minimal disclosure* and is the key privacy guarantee during credential presentation.

The security properties formalized above—completeness, consistency, attribute-based indistinguishability, and predicate-based indistinguishability—establish the core privacy and security guarantees of DIMSEPP against semi-honest adversaries in a single protocol execution. These guarantees ensure that sensitive attribute values remain confidential and that credential presentations reveal no information beyond the validity of the requested predicate. Properties related to unlinkability across multiple sessions are beyond the scope of this foundational work.

III. FROM MODEL TO INSTANTIATION: A CONCRETE CRYPTOGRAPHIC SCHEME

In this section, we present the concrete cryptographic instantiation of the system model presented in Section II-B. While we reuse the name **DIMSEPP** to refer to this complete construction, the focus here shifts from the abstract architectural description to the precise cryptographic protocols that implement it. This instantiation provides a specific implementation of the abstract algorithms and interfaces using well-established cryptographic primitives. The following establishes the correspondence between the abstract and concrete layers:

- The abstract **Protect** operation is implemented using a **Pedersen commitment** scheme to protect the attribute value.
- The **VerifyOpen** operation, which verifies the knowledge of the auxiliary value used in the commitment, corresponds to a **non-interactive Schnorr proof** of knowledge.
- For service access, the generation of the predicate proof π_θ in **ApplyServ** and its verification in **VerifyServ** are implemented as follows:
 - For inequality predicates (e.g., $v \geq \theta.\text{min}$), we employ a **ring signature-based range proof (RingProof)**.
 - For equality predicates ($v = \theta.v$), we use a **Schnorr proof**.

This concrete construction ensures that the system meets the security properties defined in Section II-C while maintaining practical efficiency. The following subsections detail the mathematical foundations and the resulting algorithmic implementations.

A. Mathematical Background

1) *Pedersen commitment*: Let $\mathbb{G}$ be a cyclic subgroup of an elliptic curve $E(\mathbb{F}_p)$ of prime order q , with base points $G, H \in \mathbb{G}$ where H is a random point with unknown discrete

log with respect to G . The commitment [26] to a secret value $v \in \mathbb{Z}_q$ with random number $r \in \mathbb{Z}_q$ is:

$$C(v, r) = [v]G + [r]H \quad (1)$$

where $[n]P$ denotes scalar multiplication on $E(\mathbb{F}_p)$.

Key Properties

- **Perfect Hiding** For any $v' \in \mathbb{Z}_q$, there exists $r' \in \mathbb{Z}_q$ such that $C(v, r) = C(v', r')$. Thus, $C(v, r)$ reveals no information about v .
- **Computational Binding** Under the **Elliptic Curve Discrete Logarithm Problem (ECDLP)** assumption, no PPT adversary can find $(v', r') \neq (v, r)$ satisfying:

$$[v]G + [r]H = [v']G + [r']H. \quad (2)$$

2) *Non-Interactive Schnorr Proof (Schnorr)*: Let $\mathbb{G}$ be a cyclic subgroup of an elliptic curve $E(\mathbb{F}_p)$ with prime order q and generator G . The non-interactive Schnorr proof [32], [33] allows a prover $\mathcal{P}$ to demonstrate knowledge of a secret $x \in \mathbb{Z}_q$ for a public key $Y = [x]G$ without interaction. This mechanism involves two cryptographic operations: Prove (executed by $\mathcal{P}$ to generate proofs) and Verify (performed by verifier $\mathcal{V}$ to validate proofs without accessing sensitive data). Prove(Y, G, x) $\rightarrow \pi$:

- 1) $\mathcal{P}$ selects $r \xleftarrow{\$} \mathbb{Z}_q$ and computes $R = [r]G$.
- 2) Computes $c = \mathcal{H}(R \parallel Y)$, where $\mathcal{H} : \{0, 1\}^* \rightarrow \mathbb{Z}_q$ is a cryptographic hash function.
- 3) Computes $s = (r + c \cdot x) \bmod q$.
- 4) Outputs the proof $\pi = (R, s)$.

Verify(Y, G, π) $\rightarrow b, b \in \{0, 1\}$:

- 1) $\mathcal{V}$ recomputes $c = \mathcal{H}(R \parallel Y)$.
- 2) if $[s]G \stackrel{?}{=} R + [c]Y$, outputs 1; else outputs 0.

Key Properties:

- **Unforgeability**: Secure against existential forgery under chosen-message attacks (EUF-CMA) in the random oracle model, assuming the hardness of the discrete logarithm problem in $\mathbb{G}$.
- **Zero-Knowledge**: The protocol is simulatable via:
 - 1) Choose random $c^* \xleftarrow{\$} \mathbb{Z}_q$ and $s^* \xleftarrow{\$} \mathbb{Z}_q$
 - 2) Compute $R^* = [s^*]G - [c^*]Y$
 - 3) Program the random oracle to satisfy $\mathcal{H}(R^* \parallel Y) := c^*$
The simulation is perfect since (R^*, s^*) is identically distributed to real transcripts.
- **Size**: The proof π consists of one curve point (R) and one scalar (s), totaling $\lceil \log_2 q \rceil + |\mathbb{G}|$ bits.

3) *Ring Signatures (RingSig)*: A ring signature scheme [34], [35], [36] allows a signer to anonymously sign a message on behalf of a *ring* (set) of public keys. Formally, given:

- A ring of n public keys: $R_{PK} = \{PK_1, \dots, PK_n\}$
- Signer’s secret key sk_s corresponding to $PK_s \in R_{PK}$
- Message $m \in \{0, 1\}^*$

The scheme consists of algorithms:

$$\sigma \leftarrow \text{Sign}(sk_s, R_{PK}, m) \quad (3)$$

$$\{0, 1\} \leftarrow \text{Verify}(R_{PK}, m, \sigma) \quad (4)$$

Key Properties:

- **Anonymity:** For any $PK_i \in R_{PK}$, the probability that an adversary identifies s is at most $1/n + \text{negl}(\lambda)$.
- **Unforgeability:** No PPT adversary can produce a valid σ without knowing any sk_i for $PK_i \in R_{PK}$, under the Discrete Logarithm assumption.

4) *Ring Signature-based Range Proof (RingProof):* The scheme[31] enables a prover to demonstrate that a committed value v lies within the range $[0, 2^n - 1]$ without revealing v or the random number r . This is achieved by combining Pedersen commitments with ring signatures for each bit of v .

The scheme consists of two main phases: Prove and Verify.

- **Prove(v, r) $\rightarrow \pi$,** where $v \in [0, 2^n - 1]$, $r \in \mathbb{Z}_q$, $C = [v]G + [r]H$:
 - 1) Bit decomposition and random number allocation:
 - Represent v in binary: $v = \sum_{j=0}^{n-1} b_j 2^j$ where $b_j \in \{0, 1\}$ for $j \in \{0, 1, \dots, n-1\}$
 - For $j \in \{0, 1, \dots, n-2\}$, sample $r_j \xleftarrow{\$} \mathbb{Z}_q$
 - Compute $r_{n-1} = r - \sum_{j=0}^{n-2} r_j \bmod q$
 - 2) Ring construction for each bit:

$$PK_{j,0} = [r_j]H + [b_j 2^j]G \quad (5)$$

$$PK_{j,1} = PK_{j,0} - [2^j]G \quad (6)$$

$$R_j = \{PK_{j,0}, PK_{j,1}\} \quad (7)$$

- 3) Ring signature generation:

$$\sigma_j \leftarrow \text{RingSig.Sign}(r_j, R_j, PK_{j,0}) \quad (8)$$

where r_j is the secret key for PK_{j,b_j} , $PK_{j,0}$ is used as message to sign. Return $\pi = \{\sigma_j\}_{j=0}^{n-1}$

- **Verify(C, π) $\rightarrow b$,** where $b \in \{0, 1\}$:
 - 1) Verify all ring signatures: $\text{RingSig.Verify}(R_j, PK_{j,0}, \sigma_j) = 1$ for all $j \in \{0, 1, \dots, n-1\}$; else return 0.
 - 2) Check commitment consistency: $\sum_{j=0}^{n-1} PK_{j,0} \stackrel{?}{=} C$; else return 0.
 - 3) Return 1 if all checks pass.

Key Properties:

- **Perfect Privacy:** The proof reveals nothing beyond $v \in [0, 2^n - 1]$
- **Computational Soundness:** Forging a proof requires solving the ECDLP or breaking the ring signature scheme
- **Compactness:** The proof size is $O(n)$, linear in the bit-length n . Specifically, π comprises n ring signatures $\{\sigma_j\}$, each of which has size determined by the particular ring signature scheme instantiation.

B. Implementing the Core Algorithms

The abstract algorithms defined in Section II-B are realized through the following four-phase operational structure. This structure implements the system model depicted in Fig. 1 using the cryptographic primitives introduced in Section III-A. The six core algorithms are distributed across these phases as follows: the **Setup** algorithm constitutes the Initial Phase; the **CreateDIDD** algorithm corresponds to the Registration Phase; the **ApplyCred** and **GenCred** algorithms are executed during the Credential Issuance Phase; and the **ApplyServ** and **VerifyServ** algorithms form the core of the Service Access Phase.

1) *Initial Phase:* The **Setup** algorithm is executed by a party to generate public parameters (pp) for the system. This process requires **no trust** in this party as it involves **no secret values**. The only requirement is that the discrete logarithm relationship between generators G and H remains unknown. This is guaranteed by deriving H from a predefined input seed (e.g., a constant string “DIMSEPP-H”) using a public **hash-to-curve** function, making the entire process transparent and publicly verifiable.

Setup(1^λ) $\rightarrow pp$: Outputs public parameters pp including:

- Elliptic curve group $\mathbb{G}$ with generators (G, H) .
- A cryptographically secure hash function $\mathcal{H} : \{0, 1\}^* \rightarrow \mathbb{Z}_q$.

2) *Registration Phase:* The registration phase implements the **CreateDIDD** algorithm and follows standard DID operations.

CreateDIDD(pp) $\rightarrow (didd, pk, sk)$:

- 1) Generate a key pair $(sk, pk) = (d, [d]G)$.
- 2) Construct a DID document $didd = \{\text{id} : \text{did}, \text{verificationMethod} : [pk], \text{service} : \dots\}$.
- 3) Return $(didd, pk, sk)$.

The user executes **CreateDIDD** to obtain $(didd, pk, sk)$, stores (pk, sk) locally, and submits $didd$ to the blockchain. The blockchain verifies $didd$'s integrity and stores it on-chain after consensus validation.

3) *Credential Issuance Phase:* This phase implements the **ApplyCred** and **GenCred** algorithms. When a user needs a verifiable credential for an attribute $(attr, v)$, they proceed as follows:

- 1) Choose a random $r \xleftarrow{\$} \mathbb{Z}_q$.
- 2) Execute **ApplyCred** to obtain (C, π_r) .
- 3) Store $(didd, attr, v, r, C)$ locally.
- 4) Submit $(didd, attr, v, C, \pi_r, \text{Proof}_v)$ to the Identity Provider (IdP), where Proof_v is externally verifiable evidence for the attribute v .

ApplyCred(pp, v, r) $\rightarrow (C, \pi_r)$:

- 1) Compute the Pedersen commitment: $C = [v]G + [r]H$.
- 2) Produce a Schnorr ZKP: $\pi_r \leftarrow \text{Schnorr.Prove}(C - [v]G, H, r)$.
- 3) Return (C, π_r) .

Example for a birth date attribute:

- Let $attr = \text{birth_date}$, $v = 19970901$.

- **Proof_v** includes a complete ePassport Data Structure (EDS) and its digital signature σ_{EDS} .

Upon receiving the tuple from the user, the IdP first verifies that *didd* is a valid and unrevoked DID document on the blockchain. It then executes **GenCred**.

GenCred(*pp*, *didd*, *attr*, *v*, *C*, π_r , **Proof_v**) $\rightarrow VC$:

- 1) Check **VerifyEvidence**(**Proof_v**) = 1; if not, return $\perp$. (For a birth date, this involves verifying the EDS and σ_{EDS}).
- 2) Check **Schnorr.Verify**($C - [v]G, H, \pi_r$) = 1; if not, return $\perp$.
- 3) Generate the verifiable credential $VC = \{\text{id} : \text{didd.id}, \text{type} : \text{attr}, \text{value} : C, \text{issuerSig} : \sigma_{\text{issuer}}, \dots\}$.
- 4) Return *VC*.

The IdP subsequently submits the *VC* to the blockchain system.

4) **Service Access Phase**: This phase implements the **ApplyServ** and **VerifyServ** algorithms. The service access predicate θ supports three forms: lower-bound ($v \geq \theta.\text{min}$), upper-bound ($v \leq \theta.\text{max}$), and equality ($v = \theta.v$).

To access a service:

- 1) The user retrieves (*didd*, $\theta.\text{attr}$, *v*, *r*, *C*) from local storage.
- 2) Verifies that *v* satisfies θ .
- 3) If compliant, executes **ApplyServ** to obtain π_θ .
- 4) Submits (*didd*, θ , π_θ) to the Service Provider (SP).

ApplyServ(*pp*, *v*, *r*, θ) $\rightarrow \pi_\theta$:

- 1) If θ is of the form $v \geq \theta.\text{min}$:
 - Generate the range proof $\pi_\theta := \text{RingProof.Prove}(v - \theta.\text{min}, r)$.
 - Return π_θ .
- 2) If θ is of the form $v \leq \theta.\text{max}$:
 - Generate the range proof $\pi_\theta := \text{RingProof.Prove}(\theta.\text{max} - v, -r)$.
 - Return π_θ .
- 3) If θ is of the form $v = \theta.v$:
 - Generate the Schnorr proof $\pi_\theta := \text{Schnorr.Prove}(rH, H, r)$.
 - Return π_θ .

VerifyServ(*pp*, *C*, θ , π_θ) $\rightarrow b_\theta$:

- 1) If θ is of the form $v \geq \theta.\text{min}$:
 - Compute $b_\theta = \text{RingProof.Verify}(C - [\theta.\text{min}]G, \pi_\theta)$.
 - Return b_θ .
- 2) If θ is of the form $v \leq \theta.\text{max}$:
 - Compute $b_\theta = \text{RingProof.Verify}([\theta.\text{max}]G - C, \pi_\theta)$.
 - Return b_θ .
- 3) If θ is of the form $v = \theta.v$:
 - Compute $b_\theta = \text{Schnorr.Verify}(C - [\theta.v]G, H, \pi_\theta)$.
 - Return b_θ .

Upon receiving an application from the user, the SP retrieves the commitment *C* from the corresponding *VC* on the

blockchain via (*didd*, $\theta.\text{attr}$). The SP then executes **VerifyServ**. If $b_\theta = 1$, access is granted; otherwise, it is rejected.

For constraints requiring a conjunction ($\theta.\text{min} \leq v \leq \theta.\text{max}$), the user must provide proofs for both predicates, and the SP grants access only if both verify successfully.

IV. SECURITY ANALYSIS OF THE PROPOSED SCHEME

This section provides a formal security analysis of the cryptographic scheme presented in Section III. We demonstrate that our construction provably satisfies the security properties defined in Section II-C—completeness, consistency, attribute-based indistinguishability, and predicate-based indistinguishability—under standard cryptographic assumptions.

A. Proof of Completeness

Theorem 1. The scheme Π described in Section III satisfies *completeness* (Definition 1). For any PPT ledger sampler $\mathcal{S}$ and security parameter $\lambda \in \mathbb{N}$, the incompleteness advantages $\text{Adv}_{\Pi, \mathcal{S}}^{\text{VC-COMP}}$ and $\text{Adv}_{\Pi, \mathcal{S}}^{\text{SERV-COMP}}$ are negligible in λ .

Proof. We analyze both experiments in Definition 1.

1. VC Generation Completeness:

- **ApplyCred** outputs (*C*, π_r) where $C = [v]G + [r]H$ and π_r is a valid Schnorr proof for the opening *r*.
- **GenCred** succeeds if **Proof_v** and π_r verify, which they do by the correctness of the external evidence verification and the Schnorr proof system.
- Thus, $\text{Adv}_{\Pi, \mathcal{S}}^{\text{VC-COMP}} = \text{negl}(\lambda)$.

2. Service Access Completeness:

- For $\theta := (v \geq \theta.\text{min})$ or $\theta := (v \leq \theta.\text{max})$, **ApplyServ** generates a range proof π_θ for $v - \theta.\text{min} \in [0, 2^n - 1]$ or $\theta.\text{max} - v \in [0, 2^n - 1]$, which verifies by the completeness of the RingProof system.
- For $\theta := (v = \theta.v)$, the Schnorr proof π_θ verifies by the completeness of the Schnorr proof system.
- Thus, $\text{Adv}_{\Pi, \mathcal{S}}^{\text{SERV-COMP}} = \text{negl}(\lambda)$. □

B. Proof of Consistency

Theorem 2. Under the *Elliptic Curve Discrete Logarithm Problem (ECDLP)* assumption, the scheme Π satisfies *consistency* as defined in Definition 2.1.

Proof. We prove consistency by contradiction. Assume there exists a PPT adversary $\mathcal{A}$ that breaks consistency with non-negligible advantage $\text{Adv}_{\Pi, \mathcal{A}}^{\text{CONS}}(\lambda)$. We construct an algorithm $\mathcal{B}$ that uses $\mathcal{A}$ to solve the ECDLP.

- 1) $\mathcal{B}$ receives an ECDLP instance (*G*, *H*), where $H = [x]G$ for an unknown $x \in \mathbb{Z}_q$.
- 2) $\mathcal{B}$ simulates the scheme's environment for $\mathcal{A}$:
 - Generates public parameters $pp = (\mathbb{G}, G, H, \mathcal{H})$.
 - Provides $\mathcal{A}$ with pp and oracle access to $\mathcal{O}^{\text{DIMSEPP}}$.
- 3) $\mathcal{A}$ outputs $(v^*, v'^*, r^*, r'^*, C^*, \pi_r^*)$ such that: $C^* = [v'^*]G + [r'^*]H$, and π_r^* verifies $C^* - [v^*]G = [r^*]H$.
- 4) From the verification equation:

$$[v'^* - v^*]G = [r^* - r'^*]H.$$

Since $v'^* \neq v^*$, $\mathcal{B}$ computes:

$$x = \frac{v'^* - v^*}{r'^* - r^*} \mod q,$$

solving the ECDLP for $H = [x]G$.

If $\mathcal{A}$ succeeds with non-negligible probability, $\mathcal{B}$ solves ECDLP with the same probability, contradicting the ECDLP assumption. Therefore, $\text{Adv}_{\Pi, \mathcal{A}}^{\text{CONS}}(\lambda)$ must be negligible. Thus, the scheme Π satisfies consistency under the ECDLP assumption. $\square$

C. Proof of Attribute-Based Indistinguishability

Theorem 3. The scheme Π satisfies *attribute-based indistinguishability* (Definition 2.2) under the computational hiding property of Pedersen commitments.

Proof. We prove attribute-based indistinguishability by constructing a reduction to the hiding property of Pedersen commitments. Assume a PPT adversary $\mathcal{A}$ breaks it with advantage $\text{Adv}_{\Pi, \mathcal{A}}^{\text{AIND}}(\lambda)$. We build an algorithm $\mathcal{B}$ that uses $\mathcal{A}$ to attack the hiding property.

- 1) $\mathcal{B}$ receives a Pedersen commitment challenge (G, H) where $H = [x]G$ for unknown $x \in \mathbb{Z}_q$.
- 2) $\mathcal{B}$ simulates the environment for $\mathcal{A}$:
 - Generates $pp = (\mathbb{G}, G, H, \mathcal{H})$.
 - Provides $\mathcal{A}$ with pp and oracle access to $\mathcal{O}^{\text{DIMSEPP}}$.
- 3) In the challenge phase:
 - $\mathcal{A}$ submits (v_0, v_1) with $v_0 \neq v_1$.
 - $\mathcal{B}$ forwards (v_0, v_1) to the Pedersen challenger, receiving $C_b = [v_b]G + [r]H$ for a random $b \leftarrow \{0, 1\}$.
 - $\mathcal{B}$ gives C_b to $\mathcal{A}$.
- 4) After additional queries (excluding direct queries about C_b), $\mathcal{A}$ outputs a guess b^* .
- 5) $\mathcal{B}$ outputs b^* as its solution.

The simulation is perfect because all oracle queries are answered using real operations and C_b is identical to a real commitment. $\mathcal{A}$'s advantage is preserved: $|\Pr[\mathcal{B} \text{ wins}] - \frac{1}{2}| = \text{Adv}_{\Pi, \mathcal{A}}^{\text{AIND}}(\lambda)$. A non-negligible advantage for $\mathcal{A}$ would thus break the hiding property of the Pedersen commitment. Therefore, $\text{Adv}_{\Pi, \mathcal{A}}^{\text{AIND}}(\lambda)$ must be negligible, and the scheme satisfies attribute-based indistinguishability. $\square$

D. Proof of Predicate-Based Indistinguishability

Theorem 4. The scheme Π satisfies *predicate-based indistinguishability* (Definition 2.3) under the anonymity of the underlying ring signature scheme.

Proof. We prove predicate-based indistinguishability by reduction to the anonymity of the ring signature. Assume a PPT adversary $\mathcal{A}$ breaks it with advantage $\text{Adv}_{\Pi, \mathcal{A}}^{\text{PIND}}(\lambda)$.

- 1) The challenger $\mathcal{C}$ simulates the environment for $\mathcal{A}$:
 - Generates $pp = (\mathbb{G}, G, H, \mathcal{H})$ and initializes $\mathcal{O}^{\text{DIMSEPP}}$.
 - Provides $\mathcal{A}$ with pp and oracle access.
- 2) In the challenge phase:

- $\mathcal{A}$ submits (v_0, v_1, θ) where $v_0 \neq v_1$ and both satisfy θ .
 - $\mathcal{C}$ obtains and sends (C_b, π_θ) to $\mathcal{A}$ as defined.
- 3) $\mathcal{A}$ continues queries under the restrictions and outputs a guess b^* .
 - 4) The proof leverages the structure of π_θ . For a range predicate, π_θ consists of ring signatures σ_j for each bit. If $\mathcal{A}$ can distinguish which v_b was used, it implies it can distinguish which secret key was used in at least one of the ring signatures σ_j , breaking their anonymity.

The simulation is perfect for non-challenge operations. $\mathcal{A}$'s advantage in distinguishing v_b directly translates to an advantage in breaking ring signature anonymity: $\text{Adv}_{\mathcal{A}}^{\text{RS-ANON}}(\lambda) = \text{Adv}_{\Pi, \mathcal{A}}^{\text{PIND}}(\lambda)$.

Since the ring signature scheme is anonymous, $\text{Adv}_{\Pi, \mathcal{A}}^{\text{PIND}}(\lambda)$ must be negligible. Thus, the scheme satisfies predicate-based indistinguishability. $\square$

E. Security Summary

Combining Theorems 1–4, the cryptographic scheme Π instantiated in Section III achieves *provable security* under standard cryptographic assumptions. It satisfies all security requirements defined in Section II-C: completeness, consistency, attribute-based indistinguishability, and predicate-based indistinguishability, thereby providing a formally grounded solution to the problem stated in Section II-A.

F. Discussion and Security Boundaries

The security guarantees established above hold rigorously within our formal model—specifically, for *single* protocol executions under the semi-honest adversary assumption. To properly contextualize these guarantees for practical deployment, we discuss their scope, inherent limitations, and resistance to stronger adversarial behavior.

a) Single-Execution Guarantees and Repeated Queries.:

Our predicate-based indistinguishability guarantee ensures that from a *single* proof π_θ , a verifier learns nothing beyond the truth value of predicate θ . However, if a verifier can make *repeated, adaptive queries* on the same credential (e.g., successively verifying ‘age ≥ 19 ’, ‘age ≥ 20 ’, etc.), it could eventually reconstruct the exact attribute value. This is an inherent, information-theoretic limitation of any system that truthfully answers arbitrary predicates about a secret value, not a flaw specific to our construction. In practice, legitimate service requests are defined by specific, contextually necessary predicates (e.g., ‘age ≥ 18 ’ for voting or ‘age ≥ 60 ’ for pension claims), and application-level policies can govern credential reuse to prevent privacy erosion through unusual query patterns.

b) Resistance Against Malicious Adversaries.:

Although our formal analysis adopts the semi-honest model for clarity in defining core privacy properties, our construction inherently provides strong protection against many realistic malicious behaviors:

- **Forgery Resistance:** Creating a valid credential or predicate proof without proper authorization requires breaking

the EUF-CMA security of the digital signature scheme or the soundness of the zero-knowledge proof system—both computationally infeasible under our cryptographic assumptions. Invalid submissions are detected during verification.

- **Data Integrity:** The binding property of Pedersen commitments (ensured by Consistency) prevents adversaries from associating different attribute values with the same on-chain commitment C .
- **Replay Prevention:** By incorporating session-specific nonces into the predicate proof π_θ (a straightforward extension), the protocol can prevent proof replay across different sessions.
- **Public Verifiability:** All on-chain data are publicly accessible, and the verification algorithm `VerifyServ` is publicly executable. This enables third-party audit of any disputed verification outcome.

c) *Remaining Attack Vectors and Mitigations.*: Certain attacks remain possible against fully malicious adversaries but primarily target properties *orthogonal* to our core focus on attribute confidentiality:

- **Service Denial:** A malicious service provider could unjustly reject a valid proof, affecting *availability*. This can be mitigated through reputation systems or cryptographic receipts (requiring SPs to sign verification outcomes).
- **Protocol Abortion:** A party could abort the protocol prematurely. Our non-interactive proof generation minimizes this window, and economic mechanisms (e.g., deposits) can deter such behavior.
- **Trust in Identity Providers:** Our system assumes IdPs are trusted for initial credential issuance. Compromised IdPs can violate privacy, but this represents a compromise of the system’s trust root—a concern common to all identity systems.

In summary, our scheme provides the essential cryptographic foundation for minimal disclosure in decentralized identity. Addressing availability and accountability concerns in fully malicious settings requires additional mechanisms beyond our cryptographic core. However, the public verifiability and standard compatibility of our design *create a viable basis* for developing such mechanisms in future work.

V. PERFORMANCE EVALUATION

This section evaluates the performance of the cryptographic scheme instantiated in Section III. The objective is to validate the *practical efficiency* of the core cryptographic operations, thereby demonstrating that our provably secure construction is viable for real-world deployment, particularly in resource-constrained environments.

A. Theoretical Analysis

Our comparative analysis (Table I) evaluates our scheme against two representative systems across four key dimensions that highlight its novel contributions.

Where existing systems force tradeoffs between privacy and functionality—requiring either complete attribute disclosure

TABLE I
COMPARISON OF DECENTRALIZED IDENTITY MANAGEMENT SYSTEMS

	[17]	[18]	Our Scheme
On-chain Data	Protected	Predicate	Commitment
Minimal Disclosure	No	Yes	Yes
Requires Trusted Setup	Yes	Yes	No
Compatibility	Partial	Partial	Full

([17]), or predicate leakage ([18])—our commitment-based approach uniquely preserves both verifiability and confidentiality. While [17] and [18] rely on zk-SNARKs[37], [38] that require complex trusted setup procedures - a significant drawback in decentralized environments where reliable third parties are scarce - our system combines Pedersen commitments with efficient ring-signature-based proofs that need no trusted initialization ceremony. This approach demonstrates superior computational and storage efficiency.

Crucially, our scheme achieves these advances without compromising the full W3C compatibility essential for adoption. The comparison reveals our architectural advantage: we resolve the core transparency-privacy tension through lightweight cryptography that outperforms SNARK-based alternatives in both trust assumptions and operational efficiency.

B. Experimental Results

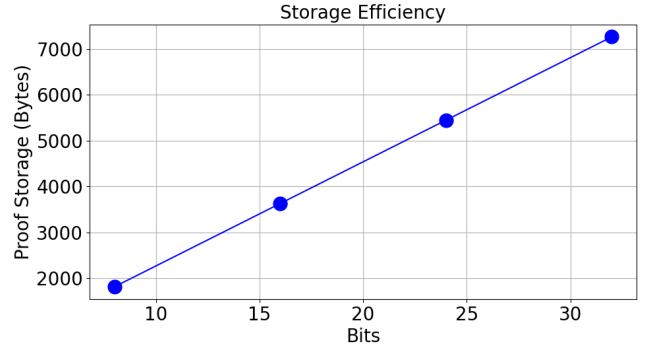


Fig. 2. storage efficiency

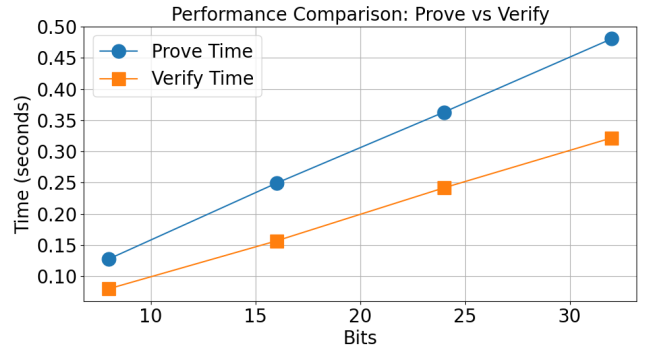


Fig. 3. computational overhead

This experimental evaluation focuses on validating the core cryptographic efficiency of the core operations in our construction—credential issuance and predicate-based service access. By implementing and measuring these algorithms in isolation, we provide fundamental, lower-bound performance data that is critical for assessing the scheme’s viability. The results establish a strong baseline for the intrinsic computational and storage overhead, which is the primary determinant of suitability for resource-constrained environments.

All experiments were conducted in a single-threaded environment to isolate baseline performance metrics and eliminate potential concurrency-related overhead. This setup ensures a clear evaluation of the cryptographic operations’ intrinsic efficiency. Note that further performance improvements can be achieved through multi-threading or parallel processing, as key operations (e.g., ring signature generation/verification for individual bits) are inherently parallelizable.

The experiments were performed on a HP laptop running Windows 11 Pro (23H2) with an Intel Core i5-10210U processor and 32GB DDR4 RAM, using a VirtualBox 7.0 virtual machine configured with 4 vCPUs and 16GB RAM running Ubuntu 20.04 LTS. The prototype system, built with Python 3.8.10, employs the starkbank-ecdsa library (v2.2.0) [39] for cryptographic operations on the secp256k1 curve. For the ring signature component required by the range proof, we utilize a variant of the algorithm from Monero [40], modified by omitting the components related to R_i (the key image) since traceability is not required in our setting. This modification is security-preserving: it removes the double-spending prevention mechanism, which is irrelevant for credential presentation, while fully retaining the anonymity and unforgeability properties that are essential for the security guarantees established in Section IV. The system consistently employs 33-byte compressed elliptic curve points, a key optimization that significantly reduces storage overhead.

For the credential issuance phase, the system demonstrates exceptional efficiency in proof generation and verification. User-side operations produce compact 65-byte Schnorr proofs in just 0.002656 seconds, while IdP verification completes in 0.005045 seconds.

For the service access phase, the system demonstrates promising results across all predicate types. For equality predicates ($v = \theta.v$), the system reuses the Schnorr proof mechanism, achieving identical efficiency to credential issuance proofs. The performance of range predicates ($v \geq \theta.min$ or $v \leq \theta.max$) is shown in Fig. 2 and Fig. 3. Fig. 2 illustrates the linear relationship between proof size and bit length (i.e., the number of bits used to represent $v - \theta.min$ or $\theta.max - v$), where each bit requires only 227 bytes. This optimization stems from our public key compression technique: since $PK_{j,1}$ can be derived directly from $PK_{j,0}$ as $PK_{j,1} = PK_{j,0} - [2^j]G$, the proof only needs to store $PK_{j,0}$. Fig. 3 demonstrates consistent computational performance, where 32-bit verification completes in 0.322 seconds while maintaining a stable 1.5× speed advantage over proof generation.

Notably, a 32-bit length suffices for most practical scenarios, supporting value differences up to 2^{32} .

The combined results from both experimental phases demonstrate that the core cryptographic operations of the proposed scheme are computationally efficient and have compact storage footprints. These findings provide a positive indication of the scheme’s *potential* suitability for practical deployment, particularly in resource-constrained environments where such efficiency is paramount. However, a comprehensive validation of this suitability requires future work on system-level integration and measurement.

VI. CONCLUSION

This paper has established a formal privacy framework for decentralized identity systems that resolves the fundamental tension between blockchain verifiability and minimal data disclosure. Through this framework, we have introduced a formal security model that precisely defines privacy requirements—including consistency, attribute-based indistinguishability, and predicate-based indistinguishability—within a semi-honest adversarial setting. To realize this model, we constructed a concrete cryptographic scheme that leverages Pedersen commitments and zero-knowledge proofs to enforce the principle of minimal disclosure, allowing users to prove predicate satisfaction without revealing sensitive attribute values.

Our construction provides rigorous, provable security guarantees under standard cryptographic assumptions while maintaining full backward compatibility with W3C DID standards. Theoretical analysis demonstrates that our approach resolves the transparency-privacy conflict inherent in blockchain-based identity systems. Furthermore, performance evaluation confirms the practical efficiency of the core cryptographic operations, with linear complexity and sub-second execution times, making the scheme suitable for resource-constrained environments such as mobile and IoT applications.

Future Work. Several directions warrant further investigation. First, optimizing proof aggregation techniques could significantly reduce communication overhead in scenarios requiring multiple predicate proofs. Second, expanding the predicate language to support more complex authorization logic would enhance the scheme’s expressiveness. Third, investigating cross-domain unlinkability mechanisms while maintaining accountability presents an important privacy-utility trade-off. Finally, comprehensive system-level integration and evaluation in production-like environments would validate the scheme’s performance at scale.

The formal framework and cryptographic instantiation presented in this work establish a foundational basis for building decentralized identity systems that are both trustworthy and privacy-preserving, aligning with evolving regulatory requirements and user expectations in an increasingly digital world.

REFERENCES

- [1] Avellaneda O, Bachmann A, Barbir A, et al. Decentralized identity: Where did it come from and where is it going?. IEEE Communications Standards Magazine, 2019, 3(4): 10-13.

- [2] Dib O, Rababah B. Decentralized identity systems: Architecture, challenges, solutions and future directions. *Annals of Emerging Technologies in Computing (AETiC)*, 2020, 4(5): 19-40.
- [3] Fdhila W, Stifter N, Kostal K, et al. Methods for decentralized identities: Evaluation and insights//International Conference on Business Process Management. Cham: Springer International Publishing, 2021: 119-135.
- [4] Stockburger L, Kokosioulis G, Mukkamala A, et al. Blockchain-enabled decentralized identity management: The case of self-sovereign identity in public transportation. *Blockchain: Research and Applications*, 2021, 2(2): 100014.
- [5] Luecking M, Fries C, Lamberti R, et al. Decentralized identity and trust management framework for Internet of Things//2020 IEEE International Conference on Blockchain and Cryptocurrency (ICBC). IEEE, 2020: 1-9.
- [6] Zhou L, Diro A, Saini A, et al. Leveraging zero knowledge proofs for blockchain-based identity sharing: A survey of advancements, challenges and opportunities. *Journal of Information Security and Applications*, 2024, 80: 103678.
- [7] Ahmed M R, Islam A K M M, Shatabda S, et al. Blockchain-based identity management system and self-sovereign identity ecosystem: A comprehensive survey. *IEEE Access*, 2022, 10: 113436-113481.
- [8] Zheng Z, Xie S, Dai H N, et al. Blockchain challenges and opportunities: A survey. *International journal of web and grid services*, 2018, 14(4): 352-375.
- [9] Nofer M, Gomber P, Hinz O, et al. Blockchain. *Business & information systems engineering*, 2017, 59(3): 183-187.
- [10] Di Pierro M. What is the blockchain?. *Computing in Science & Engineering*, 2017, 19(5): 92-95.
- [11] Yaga D, Mell P, Roby N, et al. Blockchain technology overview. *arXiv preprint arXiv:1906.11078*, 2019.
- [12] Dinh T T A, Liu R, Zhang M, et al. Untangling blockchain: A data processing view of blockchain systems. *IEEE transactions on knowledge and data engineering*, 2018, 30(7): 1366-1385.
- [13] Guo H, Yu X. A survey on blockchain technology and its security. *Blockchain: research and applications*, 2022, 3(2): 100067.
- [14] Dunphy P, Petitcolas F A P. A first look at identity management schemes on the blockchain. *IEEE security & privacy*, 2018, 16(4): 20-29.
- [15] Kuperberg M. Blockchain-based identity management: A survey from the enterprise and ecosystem perspective. *IEEE Transactions on Engineering Management*, 2019, 67(4): 1008-1027.
- [16] <https://www.w3.org/TR/did-core/>
- [17] Yang X, Li W. A zero-knowledge-proof-based digital identity management scheme in blockchain. *Computers & Security*, 2020, 99: 102050.
- [18] Maram D, Malvai H, Zhang F, et al. Candid: Can-do decentralized identity with legacy compatibility, sybil-resistance, and accountability//2021 IEEE Symposium on Security and Privacy (SP). IEEE, 2021: 1348-1366.
- [19] Liu, Y., He, D., Obaidat, M.S., et al. (2021). Blockchain-based identity management systems: A review. *Journal of Network and Computer Applications*, 186, 103079.
- [20] Voigt P, Von dem Bussche A. The eu general data protection regulation (gdpr). A practical guide, 1st ed., Cham: Springer International Publishing, 2017
- [21] Zaeem R N, Barber K S. The effect of the GDPR on privacy policies: Recent progress and future promise. *ACM Transactions on Management Information Systems (TMIS)*, 2020, 12(1): 1-20.
- [22] Albrecht J P. How the GDPR will change the world. *Eur. Data Prot. L. Rev.*, 2016, 2: 287.
- [23] ISO/IEC. (2022). Information security, cybersecurity and privacy protection—Information security management systems—Requirements (ISO/IEC 27001:2022). International Organization for Standardization.
- [24] National Institute of Standards and Technology (NIST). (2020). NIST privacy framework: A tool for improving privacy through enterprise risk management (Version 1.0). U.S. Department of Commerce. <https://doi.org/10.6028/NIST.CSWP.01162020>
- [25] Yin J, Xiao Y, Feng J, et al. Didtrust: Privacy-preserving trust management for decentralized identity. *IEEE Transactions on Dependable and Secure Computing*, 2025, vol. 22, no. 3, pp. 3105-3119
- [26] Pedersen T P. Non-interactive and information-theoretic secure verifiable secret sharing//Annual international cryptology conference. Berlin, Heidelberg: Springer Berlin Heidelberg, 1991: 129-140.
- [27] Damgård I. Commitment schemes and zero-knowledge protocols//School organized by the European Educational Forum. Berlin, Heidelberg: Springer Berlin Heidelberg, 1998: 63-86.
- [28] Goldwasser S., Micali S., Rackoff C. Knowledge complexity of interactive proofs //Proceedings of the seventeenth annual ACM symposium on Theory of computing. 1985: 291-304.
- [29] Groth J. On the size of pairing-based non-interactive arguments//Annual international conference on the theory and applications of cryptographic techniques. Berlin, Heidelberg: Springer Berlin Heidelberg, 2016: 305-326.
- [30] Sun X, Yu F R, Zhang P, et al. A survey on zero-knowledge proof in blockchain. *IEEE network*, 2021, 35(4): 198-205.
- [31] Noether S, Mackenzie A. Ring confidential transactions. *Ledger*, 2016, 1: 1-18.
- [32] Hao, Feng. Schnorr non-interactive zero-knowledge proof. No. rfc8235. 2017.
- [33] Camenisch, J., Kiayias, A., Yung, M. (2009). On the Portability of Generalized Schnorr Proofs. In: Joux, A. (eds) *Advances in Cryptology - EUROCRYPT 2009*.
- [34] Bender A, Katz J, Morselli R. Ring signatures: Stronger definitions, and constructions without random oracles//Theory of Cryptography Conference. Berlin, Heidelberg: Springer Berlin Heidelberg, 2006: 60-79.
- [35] Rivest, R.L., Shamir, A., Tauman, Y. (2006). How to Leak a Secret: Theory and Applications of Ring Signatures. In: Goldreich, O., Rosenberg, A.L., Selman, A.L. (eds) *Theoretical Computer Science. Lecture Notes in Computer Science*, vol 3895. Springer, Berlin, Heidelberg.
- [36] Wang L, Zhang G, Ma C. A survey of ring signature. *Frontiers of Electrical and Electronic Engineering in China*, 2008, 3(1): 10-19.
- [37] Groth J, Maller M. Snarky signatures: Minimal signatures of knowledge from simulation-extractable SNARKs//Annual International Cryptology Conference. Cham: Springer International Publishing, 2017: 581-612.
- [38] Eberhardt J, Tai S. Zokrates-scalable privacy-preserving off-chain computations//2018 IEEE International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data (SmartData). IEEE, 2018: 1084-1091.
- [39] <https://github.com/starkbank/ecdsa-python>
- [40] <https://www.getmonero.org/resources/research-lab/pubs/cryptonote-whitepaper.pdf>